TARNDEEP SINGH

Computer Science & Data Science Undergraduate | B.Tech 2027

+91 8699242254 | tarndeep452@gmail.com | LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

B.Tech Computer Science and Engineering student specializing in Data Science at Akal University (CGPA: 9.02/10), with hands-on experience developing machine learning pipelines, computer vision systems, and full-stack applications. Experienced in Python, Java, data analysis, model evaluation, and integrating AI capabilities into practical applications.

TECHNICAL SKILLS

Programming: Python, Java, SQL

Machine Learning: Machine Learning, Deep Learning, Computer Vision, Classification, Feature Engineering, Data Preprocessing, Data Augmentation, Model Evaluation

Libraries: TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, OpenCV, MediaPipe, Matplotlib, Seaborn, Plotly

Development Tools: HTML, CSS, JavaScript, Node.js, FastAPI, Flask, Streamlit, Flutter, Supabase, PostgreSQL, SQLite, Git, GitHub, Jupyter Notebook, Google Colab

PROJECTS

Indian Sign Language (ISL) Gesture-to-Text Recognition Ongoing

GitHub

- Developing a real-time recognition prototype for 50 ISL signs and one idle class using the AI4Bharat INCLUDE-50 dataset (943 videos used).
- Built a landmark-based sequence pipeline using MediaPipe Pose and Hand Landmarkers, and implemented a Conv1D-BiLSTM-Multi-Head Self-Attention model with approximately 0.88 million parameters.
- Developed inference components using FastAPI, WebSockets, and ONNX Runtime; 12 automated tests pass.

Technologies: Python, PyTorch, MediaPipe, OpenCV, FastAPI, WebSockets, ONNX Runtime

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StudySphere – Personalised AI Study Companion Dec. 2025

Live Demo | GitHub

- Developed a student productivity platform integrating subject and topic tracking, notes, deadlines, attendance, focus sessions, and habits.
- Implemented a retrieval-based AI tutor that ranks relevant passages from student notes and generates evidence-grounded answers with citations.
- Designed local-first storage with optional Supabase authentication, PostgreSQL Row Level Security, and serverless AI integration.

Technologies: JavaScript, HTML, CSS, Node.js, Supabase, PostgreSQL, Vercel Functions, PDF.js

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SONAR Rock vs. Mine Classification Sep. 2024

Live Demo | GitHub

- Developed a binary classification pipeline using 208 sonar samples and 60 frequency-band features, including data cleaning, exploratory analysis, and spectral feature engineering.
- Implemented comparison of 11+ model configurations using nested cross-validation, hyperparameter tuning, and hold-out evaluation with ROC-AUC, F1, MCC, accuracy, and mine recall.
- Built a Streamlit prediction interface with threshold-based decisions and low-confidence review flags.

Technologies: Python, Scikit-learn, Pandas, NumPy, Streamlit

EDUCATION

Akal University, Talwandi Sabo, Punjab Aug. 2023 – Jun. 2027

B.Tech – Computer Science and Engineering (Data Science) **CGPA: 9.02/10**

RELEVANT COURSEWORK

Software Engineering, Operating Systems, Computer Networks, Data Handling and Visualization, Probability and Statistics, Machine Learning, Artificial Intelligence, Deep Learning, Natural Language Processing, Cyber Security

LEADERSHIP AND VOLUNTEERING

Founding President – DataNinjas, Akal Tech Society Nov. 2024 – Dec. 2025

Helped establish a 75-member technical community and organised a quiz (48 participants), debate (25 participants), and treasure hunt (90 participants across 30 teams).

Vice President – FutureLens, Akal Tech Society Apr. 2026 – May 2026

Coordinated debate and presentation competitions with club leadership and communication teams.

Conference Volunteer – CI2A 2026, Akal University Apr. 2026

Supported session management, documentation, and coordination with faculty and researchers.

ACHIEVEMENTS

- Selected for the internal evaluation round of Smart India Hackathon (SIH) 2025.
- Secured **1st position** in a cybersecurity quiz among 10 participating teams.
- Qualified through two rounds of the CodeForge Club individual coding hackathon (2026).
- Led TechTrail '26, a campus-wide treasure hunt involving 99 participants across 33 teams.